

Mint

Tags:

Time Limit: 1 s Memory Limit: 128 MB
 Submission: 3 AC: 3 Score: 99.81

Submit Codes AI Code Tips

Description

The Royal Canadian Mint has commissioned a new series of designer coffee tables, with legs that are constructed from stacks of coins. Each table has four legs, each of which uses a different type of coin. For example, one leg might be a stack of quarters, another nickels, another loonies, and another twonies. Each leg must be exactly the same length.

Many coins are available for these tables, including foreign and special commemorative coins. Given an inventory of available coins and a desired table height, compute the lengths nearest to the desired height for which four legs of equal length may be constructed using a different coin for each leg.

Input consists of several test cases. Each case begins with two integers: $4 \leq n \leq 50$ giving the number of types of coins available, and $1 \leq t \leq 10$ giving the number of tables to be designed. n lines follow; each gives the thickness of a coin in hundredths of millimetres. t lines follow; each gives the height of a table to be designed (also in hundredths of millimetres). A line containing 0 0 follows the last test case.

For each table, output a line with two integers: the greatest leg length not exceeding the desired length, and the smallest leg length not less than the desired length.

Input

Please Input Input Here

Output

Please Input Output Here

Samples

input	Copy
<pre> 4 2 50 100 200 400 1000 2000 0 0 </pre>	
output	Copy

800 1200
2000 2000

Submit

Codes

HZNUOJ is based on HUSTOJ

--- Maintainer List ---

Server Time: 2025-8-26 15:07:11